

Date	2/11/2025
Time id	NM2025TMID07789
Project name	Calculating Family Expenses Using Service Now
Maximum mark	4 marks

PROJECT DESIGN PHASE 2

Data Flow Diagram (DFD) – Calculating Family Expenses using Service Now

Level 0 – Context Diagram

Overview:

This high-level diagram illustrates how data flows between family members and the ServiceNow-based Family Expense Management System. The system helps record, track, and analyze family expenses efficiently.

External Entities:

- Family Member/User: Enters daily, weekly, or monthly expense details such as category, amount, and date.
- System Administrator: Manages user accounts, verifies entries, and maintains the overall system.
- Financial Advisor/Parent: Reviews reports, analyzes expense trends, and plans budgets.

Main Process:

- Family Expense Management System –The central process that handles expense data entry, categorization, storage, calculations, and report generation within ServiceNow.

Data Flows:

- From Family Member → System: Expense details (amount, date, category, payment method, and description).
- From System → Financial Advisor: Expense summaries, categorized reports, and spending trend analytics.
- From Financial Advisor → System: Budget limits, feedback, or adjustments to categories.
- From System → Administrator: System activity logs, performance summaries, and data validation reports.

Level 1 – Detailed Data Flow Diagram

Process 1: Expense Record Submission

- Family members submit expense records using a custom ServiceNow form.

- Each submission creates a new record in the Expense Management Table.
- The system validates mandatory fields using ServiceNow client-side scripts.
- Successful entries trigger a server-side workflow for further processing.

Process 2: Data Classification and Storage

- ServiceNow business rules categorize expenses (e.g., groceries, rent, utilities).
- Categorized data is stored in corresponding database tables.
- ACLs (Access Control Lists) ensure only authorized users can view or edit entries.
- Data is automatically backed up within the ServiceNow instance.

Process 3: Calculation and Analysis

- Scheduled jobs calculate monthly totals per user and category.
- ServiceNow Flow Designer automates data aggregation and calculations.
- Analytical widgets display expense trends on dashboards.

Process 4: Budget Monitoring and Notifications

- Budget Monitoring and Notifications
- Users set budgets via a configuration form in ServiceNow.
- Real-time monitoring scripts check expenses against budget thresholds.
- Notifications and alerts are sent using ServiceNow's Notification Engine.
- Workflow updates budgets or requests approvals when limits exceed.

Process 5: Report Generation and Review

- Reports are generated through ServiceNow's Reporting Module.
- Users view visual dashboards (bar charts, pie charts, etc.) for each category.
- Financial Advisors or Admins review and approve monthly reports.
- Reports are exported to PDF or Excel using ServiceNow's export functions.

Data Stores

1. **Expense Database:** Stores all recorded expenses, including date, amount, category, and description.
2. **User Profile Database:** Stores user details such as name, role (e.g., parent, child), and spending limits.
3. **Category Rules Database:** Contains predefined expense categories (e.g., Food, Rent, Utilities, Entertainment) and classification rules.
4. **Expense Tracking Log:** Tracks daily and monthly expense updates and modifications.

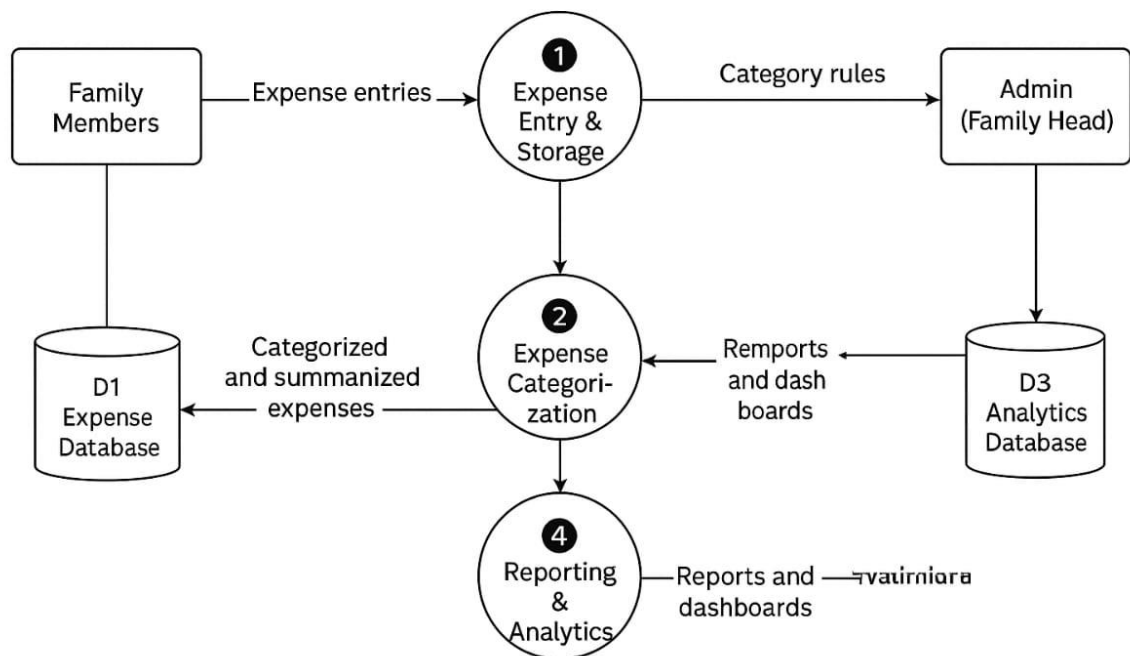
5. **Analytics Database:** Stores processed financial data for generating insights, reports, and dashboards.

Key Data Flows

- User → System: Submits expense details (amount, category, and date).
- System → User: Sends budget alerts, reminders, and expense summaries.
- System → Analytics: Transfers processed data for visualization and reporting.
- Analytics → Admin (Family Head): Displays dashboards and sends financial reports.

EXAMPLE

Family Expense Calculation System (ServiceNow)



User Stories

1. Family Member (User)

- As a family member, I want to record my daily expenses in ServiceNow so that I can track how much I spend on food, travel, and other categories easily.

2. Family Admin (Parent/Head of Family)

- As a family admin, I want to view and categorize all expenses entered by family members so that I can analyze total spending and identify where money is being overused.

3. Expense Manager (Automated Process in ServiceNow)

- As an expense manager, I want to automatically calculate the total monthly expenses and generate summary reports so that the family can plan their budget efficiently.

4. Data Analyst (Report Viewer)

- As a data analyst, I want to generate graphical reports and charts in ServiceNow that show spending trends over time, helping the family visualize and reduce unnecessary expenses.

5. Savings Planner (Goal Tracker)

- As a savings planner, I want to set savings goals and compare actual spending with planned budgets so that I can help the family achieve financial stability.

User Stories

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